

JAVED M. SHAH

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Ph.D. Candidate, Computer Science (Fall 2027) — University of Illinois Chicago • MBA, UC Berkeley Haas

RESEARCH INTERESTS

Agentic and multi-agent AI systems; game-theoretic and Bayesian intervention under uncertainty; identifiability limits in temporal coordination models; learned runtime policy for tool-using agents; and reasoning-aware defense against adaptive adversaries. Twenty-plus years scaling enterprise AI-governance and identity infrastructure ground this research in the reliability, economic incentives, and societal consequences of enterprise AI systems in production.

EDUCATION

Ph.D., Computer Science — University of Illinois Chicago

ABD; defense expected Fall 2027

Concentration: Machine Learning, Game Theory, Computational Complexity

Dissertation: Early Intervention Against Strategic Human and AI Actors — a six-paper program spanning Bayesian intervention, identifiability limits in temporal coordination models, learned runtime policy for tool-using agents, reasoning-aware adversarial defense, time-inconsistent equilibrium mediation, and the strategic effects of enforcement transparency.

MBA — Haas School of Business, University of California, Berkeley

2014

GMAT 740; Coursework emphasis: Game Theory for Decisioning and Strategy

B.S., Computer Engineering — University of Pune

2001

Computer Architecture, Artificial Intelligence, Cryptography

DOCTORAL RESEARCH

Doctoral Researcher, Department of Computer Science — University of Illinois Chicago

Advisor: Prof. Natalie Parde

- Developing a unified research program on early intervention against strategic human and AI actors, combining Bayesian inference, equilibrium design, temporal point processes, and runtime evaluation of tool-using agents.
- Designing theory and experiments together so that each formal claim is tied to a measurable deployment outcome, including attack prevention, intervention cost, obedience, identifiability, and delay.
- Earlier research examined machine-learning methods for cognitive-distortion detection and control methods for quantum systems.

PUBLICATIONS AND MANUSCRIPTS

Peer-Reviewed / Accepted

Shah, J.M. and Parde, N. Bayesian Correlated Equilibrium for Insider-Threat Detection (BTCE). GameSec 2026.
Top paper, oral presentation. <https://arxiv.org/abs/2609.03096> [Job market paper]

Under Review

Shah, J.M. and Parde, N. Identifiability and Recovery of Episodic Coordination in Switching Marked Point Processes (IDREC). Under review, Transactions on Machine Learning Research (TMLR).

Shah, J.M. and Parde, N. Online Regret-Bounded Intervention for Tool-Using Agents (ORBIT). Target venue: ICLR 2027.

Shah, J.M. and Parde, N. Process-Informed Verification of Opponent Trajectories (PIVOT). Target venue: AAMAS 2027.

Shah, J.M. and Parde, N. Time-Inconsistent Equilibrium Mediation with Persistent Obedience (TEMPO). Target venue: AAMAS 2027.

Shah, J.M. and Parde, N. Cognitive Distortion Detection with LLM-Generated Dialogue Datasets. Under review, ACL Rolling Review (ARR) 2026.

Working Papers / In Preparation

Shah, J.M. and Parde, N. Future-Incentive Limits on Transparency in Enforcement Records (FILTER). Target venue under consideration (NeurIPS 2027).

Other Publications (Interdisciplinary)

Shah, T.M., Jha, S., & Shah, J.M. (2025). Justice Denied! COVID-19 and Human Trafficking in India and the USA. Contemporary Justice Review. <https://doi.org/10.1080/10282580.2025.2452541>

Shah, J.M. & Shah, T.M. (2024). Healing Through Empathy: Machine Learning for Adaptive Therapy for Children and Youth in Armed Conflict. Emerald Publishing. <https://doi.org/10.1108/S1537-466120240000035014>

Shah, T.M. & Shah, J.M. (2024). AI's Role in Enhancing Humanitarian Efforts for Children in Armed Conflict. Emerald Publishing. <https://doi.org/10.1108/S1537-466120240000034009>

Shah, T.M. & Shah, J.M. (2024). Emotions in Politics: A Review of Contemporary Perspectives and Trends. International Political Science Abstracts, 74(1), 1–14. <https://doi.org/10.1177/00208345241232769>

Industry Whitepapers

The Congruence Model for Security. (2021)

Containing Third-Party Identity Risk. (2020)

Proposed Implementation Framework for Automatically Generating XACML Policies from Metadata. (2019)

Delegated and Chained Authorization with OAuth2 and User-Managed Access. (2018)

Modeling User Behavior Using ForgeRock and Guardian Analytics. (2017)

SELECTED RESEARCH CONTRIBUTIONS

Bayesian-Temporal Correlated Equilibrium (BTCE)

- Introduces equilibrium-mediated intervention using dynamic Bayesian evidence and correlated recommendations for strategic, present-biased users.

Identifiability and Recovery of Episodic Coordination (IDREC)

- Studies when exactly episodic coordination can be identified and recovered in switching marked point processes, including weak-link regimes evaluated through 20-seed and 50-seed confirmatory sweeps.

Online Regret-Bounded Intervention for Tool-Using Agents (ORBIT)

- Develops a learned finite-state mediator with ALLOW, INSPECT, and BLOCK actions; persistent Bayesian belief over compromise and active deviation; strategic response modeling; and an exact empirical linear program that constrains profitable deviations.

Cognitive Distortion Detection (COD)

- Develops an LLM+RLHF pipeline that generates synthetic therapist–patient dialogues with labeled cognitive distortions, outperforming BERT-based classifiers on public and synthetic benchmarks.

Applied Systems: T-RAG and LangGraph Accelerators

- Built T-RAG, a knowledge-graph-constrained retrieval system for non-human-identity investigation that retrieves directed evidence chains over time for auditable, time-aware outputs.
- Developed production LangGraph accelerators for non-human-identity workflows, integrating agent memory, vector retrieval, and enterprise API connectivity.

Extensions to Adaptive Adversaries and Enforcement Design

- Extends the dissertation program toward process-informed adversarial search, persistent obedience under time inconsistency, and the future incentive effects of transparent enforcement records.

HONORS AND RECOGNITION

- Top Paper, GameSec 2026, for Bayesian Correlated Equilibrium for Insider-Threat Detection.
- Selected for oral presentation, GameSec 2026.
- Rainmaker of the Year, 2005 and 2006.

SELECTED PRESENTATIONS

Academic

- Bayesian Correlated Equilibrium for Insider-Threat Detection. GameSec 2026, oral presentation.
- Applications and Societal Implications of AI Systems. Invited colloquium, Department of Sociology, Ohio University, 2026.

Professional & Executive Education

- “Identity for Agentic AI: Product Strategy for 2030” — Peller Tech Summit, 2025 (sponsored by BCBS RI)
- “ROI Operating Model for Identity Verification” — Peller Tech Summit, 2026 (sponsored by Brown University Health)
- “Secure Passwordless Authentication” — Gartner IAM Summit, 2023
- “OAuth 2.0 Token Exchange for Microservices” — ForgeRock Identity Summit, 2019

TEACHING EXPERIENCE

Adjunct Faculty — Computer Science

University of Illinois Chicago — Summer 2027 (planned)

- Summer course instructor for Department of Computer Science.

Adjunct Faculty — Computer Science

Ohio University — 2025–Present

- Designed and delivered AI curriculum for sociology courses, translating agentic AI, governance, and societal-impact concepts for a non-technical, interdisciplinary audience.
- Connected AI systems research to social-justice and human-trafficking topics, drawing on joint published work with Prof. Shah.

Workshop Facilitator — Enterprise Identity, Fraud & AI Governance

2012–Present

- 15 years as a workshop facilitator, designing and leading 5-10 day intensive professional-education workshops for enterprise technical and executive audiences, including DBS Bank of Singapore, Audi in Ingolstadt Germany, Blue Cross Blue Shield of North Carolina and Blue Cross Blue Shield of Rhode Island.
- Developed curricula spanning identity governance, fraud-risk strategy, architecture, and applied AI/security decision-making for cross-functional teams.

COURSES PREPARED TO TEACH

Primary: Introduction to Machine Learning and Artificial Intelligence · Introduction to Game Theory · Cybersecurity · Introduction to Data Science

Additional: Introduction to Multi-Agent Systems · Applied Data Science for Business (capstone elective)

Proposed elective: Machine Learning & Cybersecurity for Agentic AI Systems — an emerging-topics elective for advanced undergraduates, drawing directly on dissertation research in adversarial agent behavior, runtime intervention policy, and reasoning-aware defense.

PROFESSIONAL APPOINTMENTS

Managing Principal, Product & AI Platforms — Peller Technologies

Nov 2023–Present

- Built and led advisory programs supporting a \$150M+ cross-industry portfolio, growing annual pipeline 10x.

- Closed \$10M+ in new advisory engagements while defining platform roadmaps and AI-governance frameworks for Fortune 500 clients in banking and healthcare.

Vice President of Product Management — 1Kosmos Inc.

Jun 2021–Nov 2023

- Owned full product strategy for BlockID, a 50M-user decentralized identity platform supporting secure multi-factor authentication aligned to NIST SP 800-63-3 (IAL2); built the platform's KYC and CIAM engine on OAuth 2.0 and OpenID Connect.
- Reduced fraud losses by \$2.6M for a major retail customer across 500 locations.
- Designed developer.1kosmos.com and led adoption efforts that grew active developer accounts 233% (150 to 500); launched AdminX, the platform's identity verification control plane.
- Achieved 10 global awards for best-in-class identity verification platform.
- Personally hired and managed 20 product and engineering professionals during company growth.

Senior Director of Product Management, Identity Cloud — ForgeRock (NYSE: FORG)

Dec 2013–Jun 2021

- Led the Series A–to–NYSE IPO journey for the SaaS CIAM product; bootstrapped and held global responsibility for microservices, compliance, and roadmap of ForgeRock Identity Cloud.
- Held \$42M in global P&L responsibility (\$21M for the Americas — US West, Canada, LATAM); grew regional sales from \$7M to \$50M annually; managed the Americas solution architecture team.
- First to productize delegated authorization within the OAuth2 Token Exchange protocol; personally wrote roughly 2,000 lines of code for the platform's integration perimeter with partners including Pitney Bowes and Guardian Analytics.
- As Director of Product Management, Identity Platform (2013–2016): evangelized a platform-first go-to-market strategy, designed and launched the ForgeRock Identity Microservices SKU, and led 12-Factor App compliance for containerized (Kubernetes/OpenShift) deployments, cutting customer deployment time by 60% on average.
- Helped hire or interview 40 additional product and engineering professionals.

Director of Product, North America — Prolifics

Dec 2010–Nov 2013

- Co-owned P&L and led enterprise IAM product delivery and professional services, managing a team of 11 security consultants across banking, retail, and healthcare; served as lead architect for identity and access management projects.
- Built and managed an \$11M security product portfolio for North America, supporting HIPAA-aligned identity solutions for healthcare and financial-services clients.

Chief Solutions Architect — Persistent Systems Ltd. (NSE: PERSISTENT)

Jun 2001–Jan 2010

- Designed the myaccess.ucsf.edu identity portal serving 100K+ UCSF users, integrating PeopleSoft and clinical systems via the InCommon Federation and contributing to the Shibboleth open-source project.
- Architected an IBM Tivoli identity-management rollout for Kaiser Permanente spanning 270K users across 9 regions; led the State Street Bank national IAM rollout for 1M+ customers with IBM Global Services; built biometric authentication for Payflex smartcards on point-of-sale terminals.
- One of the original developers of the enRole product (later IBM Verify Identity Governance); served as worldwide Level 3 support leader and connector developer for IBM Tivoli Identity Manager and the Access360 enRole product lines, based in India.

TECHNICAL EXPERTISE

Research methods: Bayesian inference; dynamic Bayesian networks; correlated equilibrium; linear programming; sequential decision-making; temporal point processes; no-regret learning; multi-agent systems; statistical learning; controlled simulation and ablation studies.

Software & platforms: Java, C, C#, vscode, Python; AWS; Azure; GCP; Kubernetes; Terraform; CI/CD; LangGraph; multi-tenant SaaS; microservices.

Identity & security: OAuth 2.0; OIDC; SAML; token exchange; adaptive authentication; non-human identity; agent identity.

CERTIFICATIONS

Certified Information Systems Security Professional (CISSP)

Certified Information Security Manager (CISM)

Project Management Professional (PMP)

Zachman Certified Enterprise Architect

SELECTED MEDIA & PUBLIC SCHOLARSHIP

- Author, Security Boulevard — 53 published articles on identity, security, and AI governance (2021–2023).
- Contributing author, Dark Reading.
- Featured expert/speaker profiles: RSA Conference, FIDO Alliance Authenticate Conference, TechVision Research.
- Press coverage: Yahoo Finance, Business Wire, The Paypers, Biometric Update, CityBiz.

COMMUNITY SERVICE

Court Appointed Special Advocate – Abused and Neglected Children

Contra Costa County, California — 2010–2023

- Advocacy for foster children.